Digital Object Identifier

Toward Principled Threshold Selection for Blink Region Detection in Biological Signals

Author names omitted for review

ABSTRACT In fatigue-driving studies, blink activity within the electroencephalography (EEG) is treated as marker of driver fatigue rather than noise. Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG signals. However, current approaches that use thresholding estimate a single amplitude threshold from a pool of blink-heavy and blink-free epochs concatenated together, which dilutes the amplitude statistics used for threshold estimation. This study presents an epoch-aware, three-stage pipeline that avoids this dilution by keeping blink-heavy and blink-free epochs separate when estimating the detection threshold. The three-stage pipeline first screened non-overlapping epochs by peak-to-peak amplitude, then estimated a sample-level threshold from the suspicious-epoch pool, and finally extracted blink regions and mapped them to epoch-local coordinates. The proposed pipeline and two established continuous-signal baselines were compared on the complete 32-channel montage using two naturalistic driving-EEG corpora, an internally collected dataset and public dataset, evaluated with event-level overlap matching and macro-averaged F_1. Evaluated on this 32-channel montage configuration, the proposed pipeline reached macro-F_1 scores of 88.25% on Internal and 80.68% on public dataset, significantly exceeding both baselines on both corpora after Bonferroni-corrected Wilcoxon signed-rank tests. These results indicate that epoch-aware threshold estimation yields more reliable blink detection than continuous-signal thresholding in naturalistic EEG driving recordings.

INDEX TERMS blink detection, electroencephalography, epoch-aware detection, naturalistic driving, threshold selection

I. INTRODUCTION

Blinking is a physiological process that involves the closure and reopening of the eyelids [1]. Its primary functions are to protect the ocular surface, maintain corneal moisture, and support visual comfort [1]. Beyond these physiological functions, blink behaviour has been increasingly recognised as an important indicator in fatigue-driving research [2, 3]. Variations in blink-related features, such as blink frequency, blink duration, eye-closure duration, and blink patterns, can reflect changes in alertness and cognitive state [2, 4]. As fatigue increases, drivers may exhibit slower or prolonged blinks, which can indicate reduced vigilance and a higher risk of impaired driving performance [2]. Therefore, blink behaviour provides a useful and non-invasive marker for monitoring driver fatigue [3, 5].

In the neuroscience study of fatigue driving, blink activity can be captured through external methods, such as camera-based eye monitoring, or through electrophysiological recordings, particularly electroencephalography and electrooculography [3, 5]. Although EEG is mainly used to measure brain activity, blink-related components are often present in EEG signals because eyelid movements and eye movements generate strong electrical potentials [6, 7]. These ocular activities are especially prominent in frontal EEG electrodes due to their close proximity to the eyes [6, 8]. As a result, blink information can be extracted from EEG recordings, either by analysing the ocular components embedded in the EEG signal or by combining EEG with EOG-based measurements [5, 9]. This makes EEG/EOG recordings valuable not only for studying neural activity during fatigue but also for identifying blink-related markers associated with drowsiness [2, 10].

It is important to distinguish between two major perspectives on blink activity in the existing literature [3, 6]. In many EEG studies, blink-related activity is treated as an artifact because it contaminates neural signals and may interfere with the interpretation of brain activity [6, 7]. In such cases, blink components are commonly removed during EEG preprocessing [6, 11]. However, in fatigue-driving studies, blink activity can also be treated as meaningful information rather than noise [3, 5]. From this perspective, blink events are identified, extracted, and analysed as potential indicators of driver fatigue [5, 10]. Blink detection methods can generally be classified into machine-learning-based, morphology-based, and thresholding-based approaches [9, 12]. Machine learning

methods rely on trained models to classify blink events, while morphology-based methods identify blinks based on their characteristic waveform shape [13, 14]. Thresholding-based methods, in contrast, detect candidate blink events when the signal amplitude exceeds a certain threshold value, as blink activity typically produces large-amplitude deflections, especially in frontal channels [8, 15].

Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG or EOG signals [15, 16]. The simplicity of thresholding makes it computationally efficient and easy to implement, particularly in real-time fatigue-detection systems [16, 17]. Existing approaches range from single- or few-channel detectors that apply amplitude thresholds with rule-based decision logic to hybrid pipelines that combine thresholding as an initial detection step with additional candidate-refinement mechanisms [8, 9, 12, 14–16, 18–24]. Despite these developments, selecting an appropriate threshold remains challenging because blink amplitude varies across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. Blink amplitude can vary across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. A threshold that is too low may produce false detections due to noise or other artifacts, whereas a threshold that is too high may miss genuine blink events [16, 17]. This backdrop explains why simple one-value global thresholds can be attractive, yet brittle. Therefore, adaptive and robust threshold selection remains a critical consideration for reliable blink detection, particularly in fatigue-driving applications [5, 10].

In many fatigue-driving studies, EEG recordings are segmented into epochs that correspond to specific task events or analysis windows [25]. This epoch-structured setting introduces additional challenges for threshold-based blink detection [16, 17]. When thresholds are estimated from the entire recording, they may be biased by the presence of clean epochs that contain few or no blinks, which may distort threshold estimates and suppress blink detection in epochs where blinks are more prevalent [11, 24]. As a result, naive application of thresholding in an epoch-structured context may risk underdetection of blinks, which undermines the reliability of fatigue assessments based on blink metrics [5, 10]. Addressing this challenge requires developing blink detection methods that explicitly account for the epoch structure and can adapt threshold estimation to the specific characteristics of each epoch or subset of epochs, rather than relying on a global threshold derived from the entire recording [9, 24].

This paper makes several contributions to blink detection in fatigue driving research. First, we propose a double thresholding blink detection pipeline comprising epoch screening, sample level threshold estimation, and blink region extraction. To our knowledge, this is among the first frameworks to integrate these three stages explicitly to address the challenges associated with epoch structured EEG recordings. Second, we systematically compare the proposed pipeline against baseline methods, including BLINKER concat [8] and MNE annot. To support reproducibility and facilitate further research, we also developed an automated pipeline for extracting ocular indices from EEG recordings and made it freely available as a Python toolbox called *pyblinker* (TODO). The toolbox is intended to facilitate reproducible blink analysis and encourage more consistent reporting of ocular indices in EEG based fatigue research.

The remainder of the paper is organized as follows. Section Methods and materials describes the steps and methods used to extract blinks from EEG data, as well as the datasets we used to validate the results. Section Results presents the results of running the BLINKER toolbox on a large and varied collection of EEG data. Section The BLINKER software briefly introduces the toolbox, referring readers to the more detailed online documentation and downloads. Section Visual verification of BLINKER output describes our manual video verification process and some interesting results we obtained when comparing subject behavior captured in video with blink signals captured by EEG. Section Discussion offers some concluding remarks and discusses limitations of the method.

II. METHOD AND MATERIALS

a: Processing pipeline overview.

A. Processing Pipeline Overview

The proposed method is an epoch-aware, double-thresholding pipeline that transforms a continuous multichannel EEG recording into channel-wise blink-region detections through five sequential operations. First, the EEG signals are downsampled and band-pass filtered before being divided into fixed-length, non-overlapping epochs. The proposed double-thresholding method then operates through three detection stages. Stage A performs multichannel peak-to-peak screening to identify candidate epochs containing prominent amplitude excursions. Stage B pools the samples from these candidate epochs to estimate a robust sample-level threshold independently for each channel. Stage C applies the resulting channel-specific thresholds to all valid epochs to localize blink regions and determine their epoch identifier, channel, onset time, and duration. Figure 1 summarizes the complete processing pipeline, while Stages A, B, and C are described in detail in the following subsections.

b: Preprocessing

First, the continuous recording is downsampled to a common sampling grid of 100 Hz. Standardising the sampling rate makes the amplitude thresholds and the minimum-duration gate comparable across recordings acquired on different hardware, and reduces the per-sample computational cost of the downstream stages without discarding any information needed to resolve a blink, whose spectral content lies well below the new Nyquist frequency. Second, a zero-phase band-pass filter from 1.0 Hz to 20.0 Hz is applied which is removes slow drifts and DC offsets at the low end and high-frequency muscle and line noise at the high end, isolating the low-frequency, high-amplitude deflection that characterises an

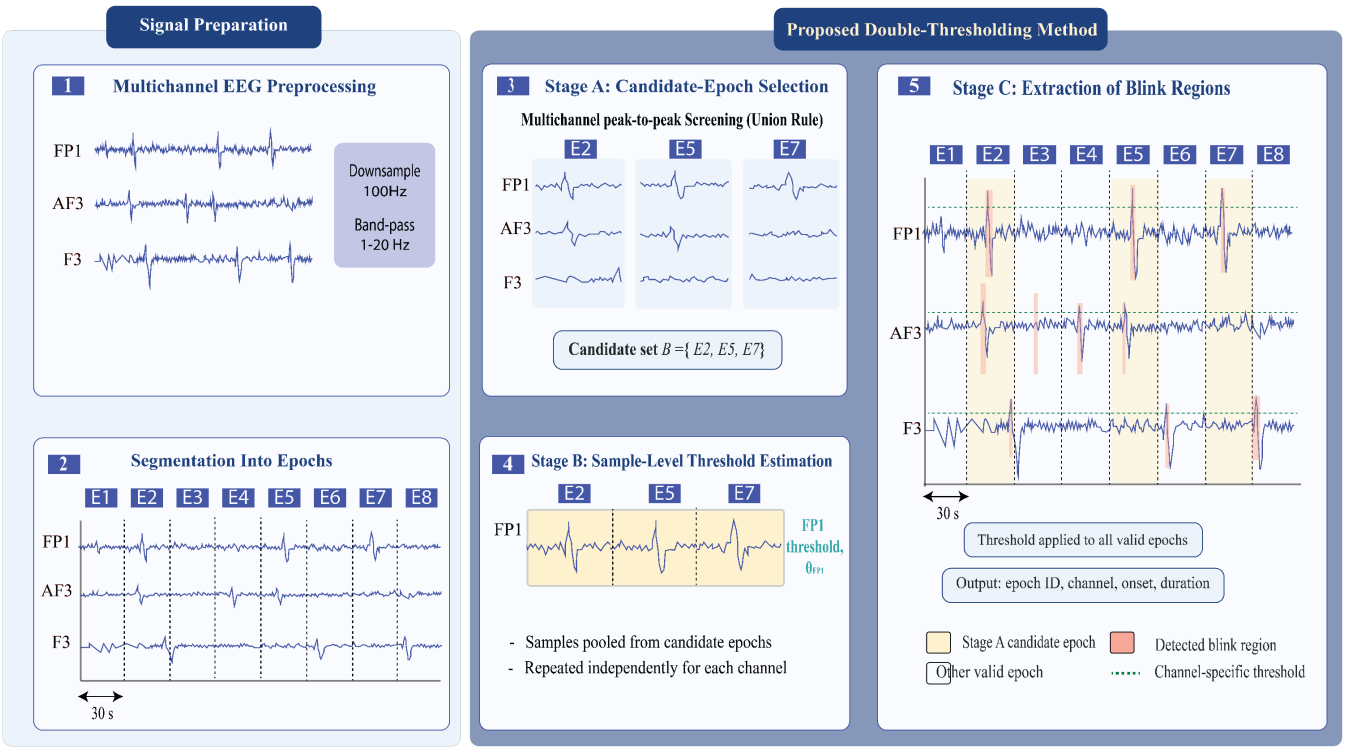


FIGURE 1. Overview of the proposed epoch-aware double-thresholding pipeline for EEG blink-region detection. Continuous multichannel EEG signals are downsampled to 100 Hz, band-pass filtered from 1 to 20 Hz, and divided into non-overlapping 30-s epochs. Stage A applies multichannel peak-to-peak screening and a union rule to select candidate epochs, represented by E2, E5, and E7 in this example. Stage B pools the samples from the selected epochs to estimate a robust sample-level threshold independently for each channel. Stage C applies the resulting channel-specific thresholds to all valid epochs and reports each detected blink region by its epoch identifier, channel, onset time, and duration. Pale yellow denotes Stage A candidate epochs, red denotes detected blink regions, and the dashed green lines represent the channel-specific thresholds.

eye blink. Third, each filtered recording is split into non-overlapping, fixed-length epochs. Epoching is what makes the thresholds *adaptive*, rather than deriving a single level for the entire session, the pipeline treats each short epoch as a local statistical context.

c: Selection of Candidate Epochs

The first stage, Stage A, of the proposed pipeline performs an initial epoch-level screening using an *autoreject*-inspired threshold-selection procedure [26] to identify epoch that are most likely to contain prominent blink-related activity. Rather than relying on a single channel, the screening decision integrates channel-level information across multiple EEG channels, allowing an epoch to be selected when abnormal amplitude excursions are detected in one or more channels. At this stage, the objective is not to localize individual blink samples, but to identify candidate epochs containing prominent amplitude activity. This multi-channel screening strategy therefore narrows the search space to potentially blink-related epochs for more detailed sample-level analysis in the subsequent detection stages.

Let $X \in \mathbb{R}^{N\times C\times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per epoch. The sample at epoch $i \in \mathcal{E} = \{1, \ldots, N\}$, channel $c \in \mathcal{C} = \{1, \ldots, C\}$, and within-epoch time index $t \in \{1, \ldots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1\le t\le T} x_{i,c,t} - \min_{1\le t\le T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c, a pattern consistent with ocular or other transient artifacts. Because signal amplitudes may vary across channels, participants, and recording conditions, a fixed peak-to-peak threshold is not imposed. Instead, a channel-specific threshold $\tau_c^\star$ is calculated from each of the channel's. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion that measures the discrepancy between retained training epochs and validation epochs.

After the channel-specific thresholds, $\tau_c^\star$, have been estimated, the peak-to-peak amplitude of each epoch is evaluated separately for each EEG channel against its corresponding threshold. An epoch is retained as a candidate when the amplitude in at least one channel exceeds the threshold $\tau_c^\star$ defined for that channel. Thus, the epoch-level screening decision incorporates information across multiple channels using a union rule, rather than relying on a single predefined

channel. The resulting set of candidate epochs is defined as The set of suspicious epochs is therefore

$$\mathcal{B} = \{i \in \mathcal{E} : b_i = 1\}.$$

where b_i denotes the epoch-level screening outcome indicating that epoch (i) satisfies the selection criterion. These candidate epochs represent high-amplitude segments that may contain prominent blink-related activity and are retained for subsequent sample-level blink-region detection.

d: Computation of Sample-Level Blink-Region Threshold Candidates

Stage B estimates channel-specific sample-level amplitude thresholds using signal samples drawn from the candidate epochs $\mathcal{B}$ identified in Stage A. The estimation is inspired by the BLINKER approach [8], combining a measure of central tendency with a robust measure of dispersion and a sensitivity parameter, allowing the threshold to adapt to the amplitude characteristics of each channel while limiting the influence of isolated extreme samples.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity.

For each channel c, all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S},\ 1 \le t \le T\}.$$

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution which can be express as below:

$$\xi_c = \mathrm{median}_{y \in \mathcal{Y}_c}\, y,$$

The dispersion is estimated by the median absolute deviation (MAD),

$$\mathrm{MAD}_c = \mathrm{median}_{y \in \mathcal{Y}_c} \left| y - \mathrm{median}_{u \in \mathcal{Y}_c}\, u \right|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\mathrm{rob}} = 1.4826\ \mathrm{MAD}_c\,.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD, and is the same scaling choice originally used in BLINKER [8].

For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k\sigma_c^{\mathrm{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

A. EXTRACTION OF BLINK REGIONS

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^{C}$ obtained in Stage B to obtain channel-wise blink regions. Specifically, blink regions are searched for within each valid epoch $i \in \mathcal{E} = \{1, \ldots, N\}$ and each channel $c \in \mathcal{C}$, by comparing the samples $x_{i,c,t}$ of that epoch and channel against the threshold candidate θ_c of the same channel.

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset time, and duration.

B. EXPERIMENTAL SETUP

The evaluation protocol was designed to compare all detector conditions on the same recordings, reference annotations, and scoring criteria. The datasets, event-level performance measures, and statistical analyses are described below.

1) Datasets

Two EEG corpora acquired during sustained simulated-driving tasks were used, providing 104 complete recording sessions in total. The corpora differed in their acquisition montages and data provenance, allowing the proposed method to be evaluated under different hardware setups.

The public corpus contains 62 sessions from 27 participants who performed an approximately 90-minute lane-keeping task in an immersive driving simulator [25], and is referred to as Cao2018 henceforth. This study used the raw recordings. All 32 recorded electrophysiological channels (30 scalp electrodes and the A1/A2 mastoid reference channels) were supplied to the pipeline. These 32 channels were Fp1, F3, Fp2, F4, FT7, FC3, C3, FC4, FT8, C4, TP7, CP3, TP8, CP4, T5, P3, O1, T6, P4, O2, F7, T3, Fz, FCz, Cz, Pz, Oz, F8, T4, CPz, A1, and A2.

The Internal corpus is a dataset collected in house, comprising 46 complete sessions recorded during a simulated lane-keeping task. EEG was acquired with a 128-channel EGI HydroCel system, with CPz as the reference. Although the original recordings used a high-density montage, the present analyses retained a predefined subset of 32 electrodes corresponding to conventional 10–20 scalp locations, following the subsampled 128-to-32-channel montage of [27]. The 32 channels are Fp1, AF3, F3, Fp2, AF4, F4, FC5, FC1, C3, FC2, FC6, C4, CP5, CP1, CP2, CP6, P5, P3, PO3, O1, PO4, P4, P7, O2, F7, T7, Fz, Cz, Pz, Oz, F8, and T8.

We divided the 32 electrodes into four anatomical regions (frontal, central, parietal, and occipital) and further divided each region into left and right hemispheres, as per [28], as shown in Figure 2. In the figure, the resulting eight regions (frontal left, frontal right, central left, central right, parietal left, parietal right, occipital left, and occipital right) are abbreviated FL, FR, CL, CR, PL, PR, OL, and OR, respectively. The figure shows the scalp locations of the 32 electrodes with the blue circles represent electrodes located in both the Internal and Cao2018 montages, the yellow circles represent electrodes located only in the Internal montage, and the red

circles represent electrodes located only in the Cao2018 montage.

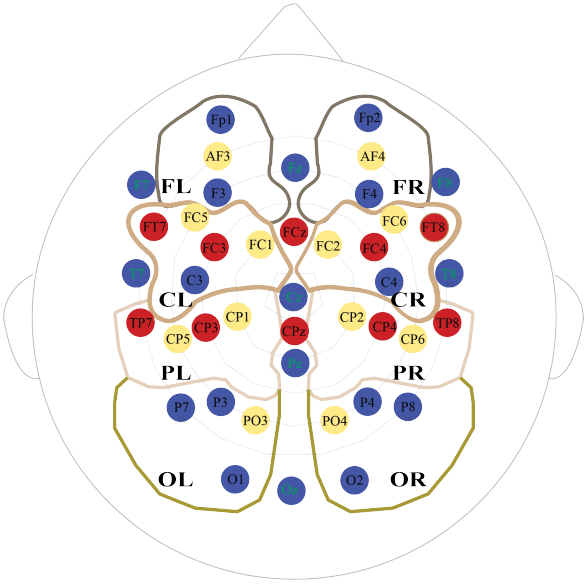


FIGURE 2. Scalp locations of the 32 electrodes used in this study, divided into eight anatomical regions (frontal left/right, central left/right, parietal left/right, and occipital left/right, abbreviated FL, FR, CL, CR, PL, PR, OL, and OR). Blue circles mark electrodes present in both the Internal and Cao2018 montages, yellow circles mark electrodes present only in the Internal montage, and red circles mark electrodes present only in the Cao2018 montage.

2) Evaluation Metrics

a: Event matching.

Performance was evaluated at the blink-event level rather than at the sample level. For a predicted interval P and a reference interval G, temporal overlap was quantified using intersection over union (IoU):

$$\mathrm{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}. \tag{1}$$

Here, $|\cdot|$ denotes interval duration. A predicted interval was eligible for a match when its IoU with an unmatched reference interval was at least 0.1. Matching was one-to-one, so a predicted or reference interval could contribute to at most one match.

b: Session-level measures.

Precision, recall, and F_1 were calculated independently for each session s from the event counts:

$$\mathrm{Precision}_s = \frac{\mathrm{TP}_s}{\mathrm{TP}_s + \mathrm{FP}_s}, \tag{2}$$

$$\mathrm{Recall}_s = \frac{\mathrm{TP}_s}{\mathrm{TP}_s + \mathrm{FN}_s}, \tag{3}$$

$$F_{1,s} = \frac{2\ \mathrm{Precision}_s\, \mathrm{Recall}_s}{\mathrm{Precision}_s + \mathrm{Recall}_s}. \tag{4}$$

c: Macro-averaging.

For a measure $M \in \{\mathrm{Precision}, \mathrm{Recall}, F_1\}$, the reported macro-average over S sessions was

$$\overline{M} = \frac{1}{S}\sum_{s=1}^{S} M_s. \tag{5}$$

Thus, each session contributed equally, irrespective of its number of annotated blinks. Pooled results across the two corpora used the same unweighted average over all 104 sessions; event counts were not pooled before computing the measures. This approach prevents sessions with many blinks from dominating the headline result, as would occur with micro-averaging.

III. RESULTS

A. EXPERIMENT 1: EEG CHANNEL-SUBSET ANALYSIS

In the proposed pipeline, Stage A screens epochs using a union rule over the channels supplied to the pipeline, so changing the channel set can also change which epochs reach Stage B. Restricting the input channels therefore changes the detector itself, not merely the signals presented to a fixed detector. The EEG channel-subset analysis accounts for this coupling by re-running the complete pipeline separately for each channel subset. We first supply the complete 32-channel montage to the pipeline as a baseline, reported in Section III-A1. This configuration assumes no prior knowledge of which brain region projects blinks most strongly, making it the most general starting point against which reduced channel sets are compared.

The analysis then evaluates anatomically defined regional subsets in Section III-A2 to identify where the EEG information most useful for blink detection is concentrated and to quantify the performance loss caused by excluding other scalp regions. Individual channels are subsequently assessed in Section III-A3 to determine whether a single optimally positioned electrode can preserve sufficient detection performance, thereby establishing the minimum channel configuration suitable for a simpler and more practical system.

1) Full-Montage Reference

a: Per-Electrode Detection

In this full-montage setup, where all 32 channels were supplied to the pipeline, per-electrode detection quality was concentrated at the frontopolar pair rather than distributed across the frontal region, as Figure 3 shows. On Internal, Fp2 and Fp1 reached F_1 values of 86.82% and 86.32%, respectively, before performance fell at the neighbouring frontal sites AF4 and AF3 to 80.32% and 73.76%, with F3, and F4 at 45.65%, and 45.23%. On Cao2018, Fp1 and Fp2 reached F_1 values of 79.68% and 78.45%, respectively, whereas the neighbouring frontal sites F3 and F4 were lower at 59.51% and 57.99%.

The central electrodes showed a dataset-specific asymmetry. On Internal, C3 reached an F_1 of 61.06% while C4 reached only 1.85%, whereas on Cao2018 C3 and C4 were

close, at 31.08% and 30.18% respectively, so the pronounced C3–C4 asymmetry was specific to Internal.

The posterior electrodes performed markedly worse than the frontal and central sites, and the parietal and occipital groups also differed from each other between datasets. On Internal, the four parietal electrodes CP1, CP2, CP5, and CP6 were uniformly near floor, with F_1 confined to a narrow 1.75%–2.19% range, and the eight occipital electrodes P7, P3, PO3, O1, PO4, E83, P4, and P8 showed only slightly more spread, from 1.77% at P4 to 3.60% at PO3. On Cao2018, the two parietal electrodes were more informative but still well below the frontal and central regions, with CP3 reaching 17.84% and CP4 reaching 19.56%. The Cao2018 occipital group was more heterogeneous than its Internal counterpart. O1 and O2 sat near floor at 3.10% and 3.33%, T5 and T6 were intermediate at 9.26% and 8.30%, and P3 and P4, the two most anterior electrodes assigned to this region, reached 12.80% and 12.67%, respectively, so the region's mean F_1 was driven up by P3 and P4 rather than by the electrodes located over true occipital cortex.

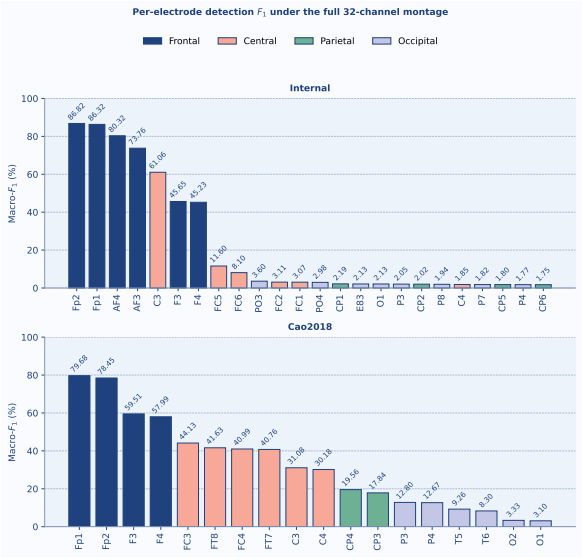


FIGURE 3. Per-electrode detection F_1 for every electrode in the four curated anatomical regions, scored inside the full 32-channel Proposed-approach montage run, coloured by scalp region. Each bar is one electrode's session-mean F_1 under the complete pipeline using all 32 channels.

2) Regional Channel Subsets

a: Per-Electrode Detection

Within these self-contained subset reruns, per-electrode detection quality remained concentrated at the frontopolar and neighbouring anterior frontal electrodes rather than distributed uniformly across the frontal region as shown in Figure 4. On Internal, within the frontal subset's own rerun, Fp2, Fp1, AF4 and AF3 were the four strongest electrodes, reaching F_1 values of 85.28%, 84.84%, 77.04% and 70.70%, respectively. Performance then fell sharply at the two remaining frontal electrodes, F3 and F4, to 42.68% and 41.00%, respectively. On Cao2018, the same four strongest electrodes, Fp1, Fp2, F3 and F4, made up the entire frontal subset, reaching F_1 values of 78.80%, 77.05%, 55.97% and 54.64%, respectively. Although every one of these frontal electrodes scored lower in its own subset rerun than the same electrode did in the full-montage run (Section III-A1), the ranking of electrodes by F_1 was unchanged from the full-montage run in both datasets, so restricting Stage A/B screening to the frontal subset alone did not reorder the strongest frontal electrodes.

The central electrodes again showed a dataset-specific asymmetry within their own subset rerun. On Internal, C3 reached an F_1 of 58.59% while C4 reached only 1.73%, so the pronounced C3–C4 asymmetry seen in the full-montage run persisted in the central subset's self-contained rerun. Wheras on Cao2018, C3 and C4 remained close together, at 27.67% and 26.62%, respectively, so this asymmetry stayed specific to Internal. Parietal and occipital electrodes were much lower in their own subset reruns. The best-performing parietal electrode was CP1 at 1.92% on Internal and CP4 at 15.63% on Cao2018, and the best-performing occipital electrode was PO3 at 3.35% on Internal and P3 at 10.04% on Cao2018. Overall, the self-contained subset reruns preserved the same anterior-to-posterior spatial pattern observed under the full montage, with detection concentrated at the frontopolar and anterior frontal sites and much lower performance at the central, parietal and occipital electrodes.



FIGURE 4. Per-electrode detection F1 within each region's own channel-subset rerun, coloured by scalp region. Each bar is one electrode scored inside that region's own self-contained subset run.

b: Electrode-Level Significance Testing

The descriptive differences between the subset reruns and the full-montage reference were next tested at the individual-electrode level using session-paired comparisons. For every electrode entering one of the four anatomical subsets shown in Figure 4, that electrode's session-level F_1 inside its own subset rerun was paired against the session-level F_1 of the

same electrode inside the full-montage run, on the sessions common to both runs. Each electrode was tested with a two-tailed Wilcoxon signed-rank test, paired by session, with p Bonferroni-corrected over every electrode tested within each dataset. Figure 5 shows, for every electrode, its session-mean F_1 under the subset rerun overlaid on the same electrode's session-mean F_1 under the full montage, together with the adjusted p-value.

On Internal, 11 of the 24 electrodes tested remained significantly different after correction, and every significant difference was a reduction in the subset rerun as shown in Figure 5A. Despite a visible drop in the descriptive means, no statistically significant difference was detected for either of the two highest-performing electrodes, Fp2 and Fp1 ($p = 1.000$ for both). But, the remaining two of the four highest-performing electrodes, AF4 and AF3, did reach significance, with reductions that were statistically significant. Among the non-frontal electrodes, C3 showed a significant reduction whereas C4 did not, so the pronounced C3–C4 asymmetry in absolute F_1 persisted regardless of whether each electrode's own subset-versus-montage change was itself significant. Several parietal and occipital electrodes also reached significance despite very low absolute F_1 as for the case for the electrode CP2, CP5, CP6, PO3 and PO4, while most others did not.

On Cao2018, significant differences were far more widespread as shown in Figure 5B. 16 of the 18 electrodes tested remained significant after correction, and again every significant difference was a reduction. As on Internal, the only electrodes not reaching significance were the frontopolar pair, Fp1 and Fp2 (adjusted $p = 1.000$ for both). The remaining two of the four highest-performing electrodes, F3 and F4, did reach significance. Unlike Internal, every central, parietal and occipital electrode tested on Cao2018 reached significance, including C3 and C4, whose reductions remained close together, consistent with the symmetry already noted between these two electrodes on this corpus.

Taken together, the electrode-level comparisons in Figure 5 show that a visibly lower descriptive mean did not by itself imply a statistically significant change. Fp1 and Fp2, the highest-performing electrode in each dataset, showed no detectable paired difference after correction in either corpus, while most other electrodes changed significantly, generally by a few percentage points. Significant differences were also more widespread on Cao2018 than on Internal, where roughly half of the non-frontal electrodes did not reach significance despite showing the same direction of change.

3) Single-Channel Operation

The regional channel-subset analysis above established how anatomical subsets of the montage behaved as self-contained detectors; the present analysis takes this reduction one step further and asks whether each individual electrode of the full montage can independently support the complete Proposed-approach detector. For every electrode retained in the full-montage configuration, the complete Stage A–

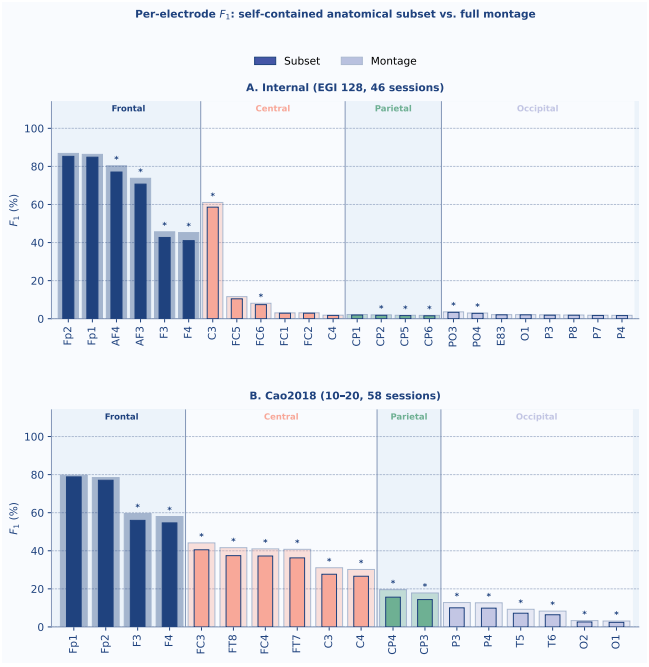


FIGURE 5. Per-electrode comparison of Proposed-approach F_1 when each electrode was evaluated inside its own self-contained anatomical subset rerun (narrow, opaque bar) against the same electrode evaluated inside the full-montage run (wide, translucent bar), for (A) Internal and (B) Cao2018. Bars show session-mean F_1; electrodes are grouped by the anatomical subset in Figure 4 and ordered by descending subset F_1 within each region. Comparisons are paired by session using two-tailed Wilcoxon signed-rank tests, with p Bonferroni-corrected separately within each dataset over every electrode tested there. An asterisk above an electrode marks adjusted $p < 0.05$; electrodes without an asterisk did not reach this threshold. Exact adjusted p-values are not printed on the bars and are instead reported in the text for electrodes discussed individually.

Stage C pipeline was rerun using that one electrode alone, with the same 30-second epochs, median centre, preprocessing and thresholding parameters as the full-montage reference, so each result is a genuine self-contained single-electrode detector rather than one electrode's score extracted from a multichannel run.

a: Per-Electrode Detection

Ranked by mean single-electrode macro F_1, Fp1, Fp2, AF4 and AF3 topped Internal's ranking, reaching F_1 values of 83.73%, 83.37%, 75.16% and 67.59%, respectively, the same four frontal electrodes identified by the earlier restricted analysis and in the same order except that Fp1 now leads Fp2 by 0.36 percentage points rather than trailing it (Figure 7; Figure 6). C3 followed next in the ranking with an F_1 of 54.05%, ahead of the remaining two canonical frontal electrodes, F4 and F3, which trailed further behind at 39.98% and 39.85%. Performance then fell sharply across the remaining 17 electrodes. The weakest was C4 with an F_1 of 1.51%, with every parietal and occipital electrode below 3.2%.

On Cao2018, Fp1, Fp2, F3 and F4 topped the single-electrode ranking, the same as Internal's canonical frontal set, reaching F_1 values of 77.65%, 75.59%, 49.30% and 49.03%, respectively, matching both the identity and the order of the

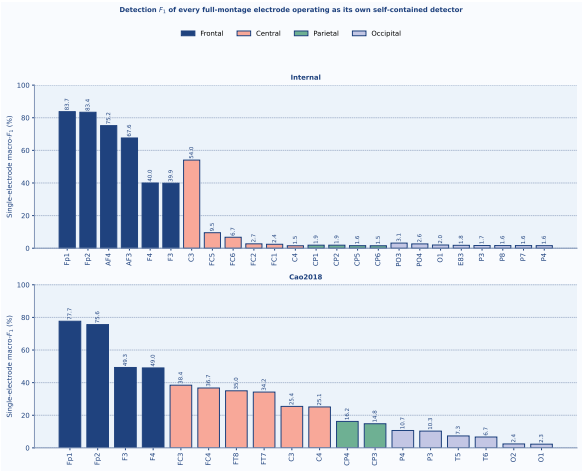


FIGURE 6. Detection F_1 of every electrode in the four curated anatomical regions operating as its own self-contained detector, in which the complete Stage A–Stage C pipeline was rerun on that one electrode alone, coloured by scalp region.

full-montage ranking (Figure 7). Performance then dropped substantially at the next electrode in the ranking, FC3, to an F_1 of 38.44%, followed by the remaining central electrodes FC4, FT8 and FT7, reaching F_1 values of 36.72%, 34.99% and 34.22%, respectively. The central region therefore occupied the entire second tier of this ranking, clearly below the four frontal electrodes and still ahead of the weaker central pair, C3 and C4, at 25.43% and 25.10%. Beyond the central electrodes, performance declined further toward parietal (14.80–16.22%) and occipital sites, with the weakest electrode, O1, at an F_1 of 2.28%. This confirms, within the four curated regions, the same anterior-to-posterior decline reported for the full-montage and regional-subset analyses (Sections III-A1 and III-A2).

b: Electrode-Level Significance Testing

The descriptive differences between single-electrode and full-montage operation were next tested for every electrode using session-paired comparisons. For each of the electrodes, that electrode's session-level F_1 operating alone was paired against the same electrode's session-level F_1 scored inside the full-montage run, on the sessions common to both. For example, Fp2 operating as the entire detector was compared with Fp2 scored inside the full-montage Proposed-approach detector, and likewise for every other electrode. Each electrode was tested with a two-tailed Wilcoxon signed-rank test, paired by session, with p Bonferroni-corrected over the electrodes tested within each dataset The Figure 7 shows, for every electrode, its session-mean F_1 operating alone overlaid on the same electrode's session-mean F_1 under the full montage, together with the adjusted-significance marker.

On Internal (Figure 7A), 14 of the 24 electrodes tested remained significantly different after correction, and every significant difference was a reduction under single-electrode operation, with ΔF_1 ranging from -7.02 to -0.15 percentage among all 24 electrodes. Despite retaining the two highest absolute single-electrode scores, neither Fp1 nor Fp2 reached significance after correction (adjusted $p = 1.000$ for both), wheras the AF4 and AF3 did reach significance which is lower in the single-electode relative to the full montage configuraion. Beyond the frontal electrodes, the Internal C3–C4 asymmetry noted above coexisted with a significant configuration effect at both electrodes, and most occipital electrodes reached significance different except O1 and P4, despite all having very low absolute F_1.

On Cao2018 (Figure 7B), 16 of the 18 electrodes tested remained significantly different after correction, again all reductions. Similarly, only the frontopolar pair, Fp1 and Fp2, did not reach significance after correction (adjusted $p = 1.000$ for both), and F3 and F4, the third- and fourth-highest single-electrode scores, showed the two largest absolute reductions ($\Delta F_1 = -10.21$ and -8.96 percentage points, respectively). Every central, parietal and occipital electrode reached significance, including both C3 and C4 (consistent with the C3/C4 symmetry already noted).

B. EXPERIMENT 2: STABILITY ACROSS EPOCH DURATIONS

Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink detector itself. In driving studies, epoch lengths ranging from a few seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F_1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

The pipeline was evaluated under epoch durations of 10, 20, 30 (reference), 40, 50, 60 and 120 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. Experiment 1 found that Fp1 and Fp2 were consistently the two best-performing electrodes on both corpora. Each electrode is thus a deployable single-electrode operating point and the epoch duration is being varied. Figure 8 reports macro-averaged F_1 for each duration, with an asterisk marking the durations whose two-tailed Wilcoxon p-value, Bonferroni-corrected over the six non-reference durations within each electrode, differs significantly from the 30-second reference.

Internal showed a less consistent, and electrode-dependent, pattern. The 30 s reference yielded each electrode's highest Internal F_1, 86.32% for Fp1 and 86.82% for Fp2. Relative to it, Fp1's F_1 fell significantly to 85.92% at 50 s ($p = 0.002$), to 85.67% at 60 s ($p < 0.001$) and to 85.04% at 120 s ($p < 0.001$). Fp2 differed significantly only at 120 s, falling to 85.57% ($p < 0.001$). Fp2's 50 s and 60 s declines, to 86.51% and 86.27% respectively, were descriptively similar to Fp1's but did not reach significance after correction, with $p = 0.080$ and $p = 0.064$ respectively. Neither electrode differed from its reference at 10, 20 or 40 s. Significance tracked the consistency of the change rather than its magnitude.

Performance on Cao2018 was stable for both electrodes, with no significant difference from the 30 s reference at any tested duration. For Fp1, the 30 s reference F_1 was 79.68%,



FIGURE 7. **Per-electrode comparison of Proposed-approach F_1 when each electrode in the four curated anatomical regions operated as a complete self-contained detector (narrow, opaque bar) against the same electrode scored inside the full 32-channel montage (wide, translucent bar), for (A) Internal and (B) Cao2018. Comparisons are paired by session using two-tailed Wilcoxon signed-rank tests, with p Bonferroni-corrected separately within each dataset . An asterisk above an electrode marks adjusted $p < 0.05$, electrodes without an asterisk did not reach this threshold.**

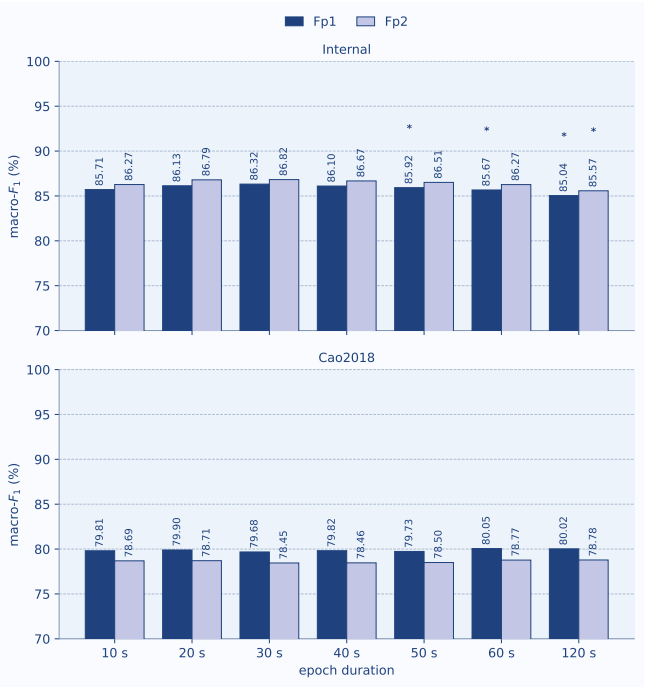


FIGURE 8. **Macro-averaged F_1 of Proposed-approach across epoch durations, on the full 32-channel montage, for Fp1 and Fp2 (Experiment 1's two consistently best-performing electrodes) reported separately: Internal (top row) and Cao2018 (bottom row). Durations of 10, 20, 30 (reference), 40, 50, 60 and 120 seconds were tested, and an asterisk marks the durations that differ significantly from the 30 s reference after Bonferroni correction.**

the lowest value across the seven durations. At the two shorter durations, the F_1 score were higher than the 30 s reference value, reaching 79.81% at 10 s ($p = 1.000$) and 79.90% at 20 s ($p = 0.381$). The F_1 is in its highest value, 80.05% at 60 s, a peak 0.38 percentage points above the reference despite not reach significance ($p = 0.882$). For Fp2, the 30 s reference F_1 was 78.45%, also its lowest value. F_1 showed the same pattern, rising to 78.69% at 10 s ($p = 1.000$) and 78.71% at 20 s ($p = 0.159$), and reaching its highest value, 78.78% at 120 s, a peak 0.33 percentage points above the reference that also did not reach significance ($p = 1.000$).

C. EXPERIMENT 3: STRATEGY COMPARISON

In driving and other epoch-structured paradigms, it is common to apply a continuous-signal blink detector to the raw or minimally preprocessed recording and subsequently map the detected blinks back onto the epoch grid. Both BLINKER and the standard MNE pipeline follow this approach, in which channel-specific thresholds are derived from a signal pool containing both blink-heavy and blink-free epochs. The central premise of the proposed pipeline is that pooling these heterogeneous epochs dilutes the amplitude statistics used for threshold estimation, thereby distorting threshold placement and shifting the resulting precision-recall trade-off.

Accordingly, we hypothesise that naive concatenation will produce systematically lower session-level F_1 scores than the epoch-aware pipeline because of a less favourable precision-recall trade-off. To evaluate this hypothesis, all three pipelines (i.e., BLINKER, MNE, and Proposed-approach) are applied to every session using 30-second epochs and all available channels. All 32 channels are included to ensure a fair comparison with BLINKER, which, by default, processes all channels provided to it and requires the user to select a representative blink channel. For consistency across methods, all pipelines are therefore supplied with the same 32 channels, and the channel yielding the highest F_1 score is selected for subsequent analysis. The macro-F_1 for Proposed-approach is summarised in Table 1, and the per-session precision-recall operating points in Figure 9.

In the strategy comparison, Proposed-approach achieved macro-F_1 scores of 88.25% and 80.68% on Internal and Cao2018, respectively, exceeding both baselines on every dataset (Table 1). Two-sided Wilcoxon signed-rank tests on session-level F_1, Bonferroni-corrected across the four baseline comparisons in that table (two baselines $\times$ two datasets), confirm that the advantage over both BLINKER and MNE is significant on both corpora ($p < 0.05$ after correction in every case). This performance gap is reflected in the precision-recall operating points shown in Figure 9, where BLINKER, consistently favors recall over precision on both corpora.

This asymmetry follows from the pooled-threshold mechanism described earlier. Deriving a single channel threshold from a mixture of blink-heavy and blink-free epochs lowers the effective detection threshold relative to an epoch-aware estimate, so more candidate events cross it, raising recall but also admitting more false positives that lower precision. This asymmetry appears as a consistent spatial ordering in the scatter, with Proposed-approach occupying the high-precision, high-recall region, BLINKER sitting in the high-recall, low-precision region, and MNE falling lower on both axes. The ordering holds not only in the pooled view but within each corpus individually, with Proposed-approach separated from both baselines in the same direction on Internal and on Cao2018.

TABLE 1. Macro-F_1 (%) for Proposed-approach and both baselines on Internal and Cao2018. An asterisk marks a baseline as significantly different from Proposed-approach (two-sided Wilcoxon signed-rank test on session-level F_1, $p < 0.05$, Bonferroni-corrected across the four tests shown).

Dataset	Proposed	BLINKER	MNE
Internal	88.25	76.54*	66.64*
Cao2018	80.68	69.40*	56.27*



FIGURE 9. Per-session precision-recall operating points for Internal and Cao2018, shown separately. Each small point is one session's precision-recall pair for one condition. Marker shape encodes the condition (BLINKER, MNE, Proposed-approach) and marker color encodes the dataset. The large navy-outlined marker for each condition marks its mean over sessions within that dataset. Both axes run from 0 to 100 percent.

IV. DISCUSSION

Experiment 1 showed that useful detection information was concentrated at the frontopolar sites and that reduced frontal configurations retained substantially more performance than other anatomical subsets. Within the full-montage run, the best individual electrode was Fp2 on Internal, reaching an F_1 of 86.82%, and Fp1 on Cao2018, reaching an F_1 of 79.68%. When the complete pipeline was instead run on one electrode alone, the best single-electrode result was obtained at Fp1 in both corpora, reaching 83.73% on Internal and 77.65% on Cao2018. Performance declined sharply for non-frontal subsets, although the subset analysis did not establish that every frontal channel was necessary. This concentration is compatible with the strong projection of blink activity to the prefrontal scalp and with earlier single-channel and forehead-EEG studies [10, 12, 15, 16, 21, 23, 29–31]. The findings therefore motivate further evaluation of reduced frontopolar systems for wearable or in-vehicle use, without establishing that such systems will reproduce the retrospective full-montage operating point in deployment.

Experiment 2 indicated that Proposed-approach, evaluated separately at Fp1 and at Fp2, the two electrodes identified in Experiment 1 as consistently the best-performing, was largely insensitive to epoch duration on Cao2018, where F_1 spread by only 0.38 percentage points at Fp1 and 0.33 percentage points at Fp2 across the seven tested durations, with no significant difference from the 30-second reference at any duration (Figure 8). This pattern is useful because epoch duration is commonly determined by the experimental paradigm rather than by the blink detector. Internal showed a less consistent, electrode-dependent pattern. Fp1's F_1 declined from 86.32% at 30 seconds to 85.92% at 50 seconds ($p = 0.002$), to 85.67% at 60 seconds ($p < 0.001$), and to 85.04% at 120 seconds ($p < 0.001$), all three remaining significant after correction, whereas Fp2 declined from 86.82% to 85.57% only at 120 seconds ($p < 0.001$), with its 50-second and 60-second drops falling short of significance ($p = 0.080$ and $p = 0.064$). Neither electrode differed from its 30-second reference at 10, 20, or 40 seconds. The evidence therefore supports stable performance from 10 to 40 seconds in both corpora for either electrode, and full stability across the complete 10–120-second range on Cao2018, but it does not establish equivalence across that full range on Internal, where the point at which performance becomes unstable is itself electrode-dependent. Longer epochs may alter the composition of the signal used during screening and threshold estimation, but the present analysis did not test that mechanism directly.

Experiment 3 showed that the epoch-aware pipeline outperformed both continuous-signal baselines on both driving corpora. Proposed-approach reached macro-F_1 scores of 88.25% on Internal and 80.68% on Cao2018, compared with 76.54% and 69.40% for BLINKER, an established blink-specific method applied to the concatenated recording [8], and 66.64% and 56.27% for MNE, the default amplitude-annotation routine of MNE-Python, with all four session-level differences remaining significant after Bonferroni correction (Table 1). The precision-recall operating points indicate how this gap arose where BLINKER consistently favoured recall over precision, operating at a more permissive point that admitted more false detections, MNE fell lower on both axes, and Proposed-approach occupied the high-precision, high-recall region in each corpus (Figure 9). This profile is consistent with the premise that a threshold estimated from a pool of blink-heavy and blink-free epochs is placed too low,

and such sensitivity to the signal pool is plausible because ocular-artifact amplitude varies across subjects, electrodes, and recording sessions [6, 8, 11, 24].

These findings are subject to several limitations. The benchmark was restricted to two driving corpora, so generalisation to other experimental paradigms or clinical populations was not tested. Event detection was reported under one overlap-based matching rule, and sensitivity to alternative event-matching criteria was not evaluated; absolute F_1 values should therefore not be compared across studies without accounting for their scoring rules [32]. Performance was not reported separately by blink duration, so detection of prolonged eye closures, which is particularly relevant to drowsiness assessment, remains unestablished. The evaluation also depends on the available reference annotations and their operational definitions of blink onset and offset, boundaries that can vary with signal representation and annotation criteria [1]. In addition, the headline strategy comparison used a retrospective best-channel-per-session rule based on reference F_1; this common oracle was applied fairly to every condition, but its scores do not represent prospective channel selection without labels. Finally, the proposed thresholding method was not benchmarked against learned detectors such as transformer-, vision-, or other deep-learning approaches [33–35]. Its demonstrated strengths are therefore limited to the methods and evaluation protocol examined here.

Future work should therefore evaluate prolonged eye closures separately, test prospective channel selection, and broaden the benchmark. Reporting normal blinks and prolonged closures independently would establish whether performance is retained for the sustained closures most relevant to drowsiness assessment [3, 5]. A prospective study should also replace the reference-based best-channel oracle with a fixed frontopolar electrode or a label-free quality criterion and quantify the resulting performance loss. Evaluation beyond driving data, including paradigms organised into task blocks, trials, or other natural segments, would test whether the observed robustness extends to different blink morphologies, cognitive demands, and recording conditions. Finally, the epoch-screening idea could be assessed as a front end to learned blink or drowsiness detectors [36–38]; such a hybrid remains a proposal and was not evaluated in the present study.

V. CONCLUSION

Taken together, the findings indicate that screening epoch-structured EEG before estimating sample-level thresholds provides more reliable event-level blink-region detection than deriving a single threshold from a concatenated recording under the evaluated driving conditions. The proposed median-based configuration outperformed the examined baselines across both driving corpora, supporting epoch screening as the principal methodological contribution of the pipeline. The spatial analysis further showed that frontopolar electrodes retained most of the full-montage performance, while the epoch-duration analysis indicated broadly stable pooled performance despite some corpus-specific sensitivity to longer epochs. Collectively, these findings support an interpretable and spatially sparse approach to epoch-aware blink detection that can accommodate common segmentation choices without requiring a dense electrode montage. The evaluation deliberately focused on driving data to establish the method's performance within its intended fatigue-driving context. Consequently, its generalizability beyond this domain and its suitability for practical deployment remain to be established. Future work should validate the pipeline across independent driving datasets and recording systems before examining its transfer to other epoch-structured paradigms. Overall, this study positions epoch screening as a practical foundation for improving blink-detection threshold calibration in heterogeneous, epoch-structured EEG.

REFERENCES

[1] M. Nyström, R. Andersson, D. Niehorster, R. Hessels, and I. Hooge, "What is a blink? classifying and characterizing blinks in eye openness signals," *Behavior Research Methods*, 2024.

[2] E. Alyan, "Blink-related eeg activity measures cognitive load during proactive and reactive driving," *Scientific Reports*, 2023.

[3] Y. Tian and J. Cao, "Fatigue driving detection based on electrooculography: a review," *Eurasip Journal on Image and Video Processing*, 2021.

[4] P. Ren, X. Ma, W. Lai, M. Zhang, S. Liu, Y. Wang, M. Li, D. Ma, Y. Dong, Y. He, and X. Xu, "Comparison of the use of blink rate and blink rate variability for mental state recognition," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2019.

[5] M. Shahbakhti, M. Beiramvand, E. Nasiri, S. Far, W. Chen, J. Sole-Casals, M. Wierzchon, A. Broniec-Wojcik, P. Augustyniak, and V. Marozas, "Fusion of eeg and eye blink analysis for detection of driver fatigue," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2023.

[6] R. J. Croft and R. J. Barry, "Removal of ocular artifact from the eeg: a review," *Neurophysiologie Clinique/Clinical Neurophysiology*, 2000.

[7] S. Akuthota, K. Kumar, and R. Janapati, "Artifacts removal techniques in eeg data for bci applications: A survey," *Computational Intelligence and Deep Learning Methods for Neuro-rehabilitation Applications*, 2023.

[8] K. K. Kleifges, N. Bigdely-Shamlo, S. E. Kerick, and K. A. Robbins, "Blinker: Automated extraction of ocular indices from eeg enabling large-scale analysis," *Frontiers in Neuroscience*, 2017.

[9] M. Agarwal and R. Sivakumar, "Blink: A fully automated unsupervised algorithm for eye-blink detection in eeg signals," 2019.

[10] L.-W. Ko, O. Komarov, W.-K. Lai, W.-G. Liang, and T.-P. Jung, "Eyeblink recognition improves fatigue prediction from single-channel forehead eeg in a realistic sustained attention task," *Journal of Neural Engineering*,

2020.
[11] A. Klein and W. Skrandies, "A reliable statistical method to detect eyeblink-artefacts from electroencephalogram data only," *Brain Topography*, 2013.
[12] Y. Zhang, X. Zheng, W. Xu, and H. Liu, "Rt-blink: A method toward real-time blink detection from single frontal eeg signal," *IEEE Sensors Journal*, 2023.
[13] M. Rashida and M. Habib, "Quantitative eeg features and machine learning classifiers for eye-blink artifact detection: A comparative study," *Neuroscience Informatics*, 2023.
[14] Y. Jiang, D. Wu, J. Cao, L. Jiang, S. Zhang, and D. Wang, "Eyeblink detection algorithm based on joint optimization of vme and morphological feature extraction," *IEEE Sensors Journal*, 2023.
[15] W.-D. Chang, H.-S. Cha, K. Kim, and C.-H. Im, "Detection of eye blink artifacts from single prefrontal channel electroencephalogram," *Computer Methods and Programs in Biomedicine*, 2016.
[16] D.-K. Tran, T.-H. Nguyen, and T.-N. Nguyen, "Detection of eeg-based eye-blinks using a thresholding algorithm," *European Journal of Engineering and Technology Research*, 2021.
[17] L. Jefri, F. Rahman, N. Malik, and F. Isa, "Eye blink region detection in eeg using amplitude thresholding," 2022.
[18] H. Nolan, R. Whelan, and R. Reilly, "Faster: Fully automated statistical thresholding for eeg artifact rejection," *Journal of Neuroscience Methods*, vol. 192, no. 1, pp. 152–162, 2010. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0165027010003894
[19] R. Ghosh, S. Phadikar, N. Deb, N. Sinha, P. Das, and E. Ghaderpour, "Automatic eyeblink and muscular artifact detection and removal from eeg signals using k-nearest neighbor classifier and long short-term memory networks," *IEEE Sensors Journal*, 2023.
[20] M. Wang, J. Wang, X. Cui, T. Wang, T. Jiang, F. Gao, and J. Cao, "Multidimensional feature optimization based eye blink detection under epileptiform discharges," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2022.
[21] G. Wang, "Sliding window higher-order cumulants for detection of eye blink artifact from short segments of single-channel eeg," *PeerJ Computer Science*, 2025.
[22] J. Cao, L. Chen, D. Hu, F. Dong, T. Jiang, W. Gao, and F. Gao, "Unsupervised eye blink artifact detection from eeg with gaussian mixture model," *IEEE Journal of Biomedical and Health Informatics*, 2021.
[23] J. Valderrama, A. De La Torre, and B. Van Dun, "An automatic algorithm for blink-artifact suppression based on iterative template matching: Application to single channel recording of cortical auditory evoked potentials," *Journal of Neural Engineering*, 2018.
[24] E. Guttmann-Flury, X. Sheng, D. Zhang, and X. Zhu, "A new algorithm for blink correction adaptive to inter- and intra-subject variability," *Computers in Biology and Medicine*, 2019.
[25] Z. Cao, C.-H. Chuang, J.-K. King, and C.-T. Lin, "Multi-channel eeg recordings during a sustained-attention driving task," *Scientific Data*, 2019.
[26] M. Jas, D. A. Engemann, Y. Bekhti, F. Raimondo, and A. Gramfort, "Autoreject: Automated artifact rejection for MEG and EEG data," *NeuroImage*, vol. 159, pp. 417–429, 2017.
[27] C. Hatlestad-Hall, R. Bruña, M. Liljeström, H. Renvall, K. Heuser, E. Taubøll, F. Maestú, and I. H. Haraldsen, "Reliable evaluation of functional connectivity and graph theory measures in source-level eeg: How many electrodes are enough?" *Clinical Neurophysiology*, vol. 150, pp. 1–16, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1388245723002286
[28] B. E. Butler and L. J. Trainor, "Sequencing the cortical processing of pitch-evoking stimuli using eeg analysis and source estimation," *Frontiers in Psychology*, vol. Volume 3 - 2012, 2012. [Online]. Available: https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2012.00180
[29] M. Shahbakhti, M. Beiramvand, I. Rejer, P. Augustyniak, A. Broniec-Wojcik, M. Wierzchon, and V. Marozas, "Simultaneous eye blink characterization and elimination from low-channel prefrontal eeg signals enhances driver drowsiness detection," *IEEE Journal of Biomedical and Health Informatics*, 2022.
[30] Y. Aoki and T. Fukami, "Drowsiness detection based on eeg and blink information measured with a single electrode," *International Journal of Innovative Computing, Information and Control*, vol. 19, no. 1, pp. 229–238, 2023.
[31] R. López-Ahumada, R. Jiménez-Naharro, and F. Gómez-Bravo, "A hardware-based configurable algorithm for eye blink signal detection using a single-channel bci headset," *Sensors*, vol. 23, no. 11, 2023.
[32] R. Salles, "Softed: Metrics for soft evaluation of time series event detection," *Computers & Industrial Engineering*, 2024.
[33] Á. Fodor, K. Fenech, and A. Lőrincz, "Blinklinmult: Transformer-based eye blink detection," *Journal of Imaging*, 2023.
[34] J. Hong, J. Shin, J. Choi, and M. Ko, "Robust eye blink detection using dual embedding video vision transformer," 2024.
[35] A. Egambaram and N. Badruddin, "An investigation to detect driver drowsiness from eye blink artifacts using deep learning models," 2022, pp. 23–29.
[36] N. Cabot, D. Lamouchi, Y. Yaddaden, and R. Cherif, "Drowsiness detection through yawning and eye blinking models using convolutional neural networks and transfer learning," 2024.
[37] F. Makhmudov, D. Turimov, M. Xamidov, F. Nazarov,

and Y.-I. Cho, "Real-time fatigue detection algorithms using machine learning for yawning and eye state," *Sensors*, 2024.

[38] B. Zhang, D. Yan, X. Wang, L. Wang, and J. Mo, "Fatigue driving detection based on fusion of eeg and blink potential features," *Journal of Railway Science and Engineering*, vol. 22, no. 8, pp. 3738–3747, 2025.